Sahaj Bhadja

(213) 284-4651 sahaj.bhadja@gmail.com sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years of experience building **cloud backend services**, **distributed systems**, and reliable **AI-assisted workflows** across Java, Python, AWS, and Azure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Achieved 99%+ reliability across 5K+ monthly jobs through **production backend pipelines**, implementing incremental processing, validation, reconciliation, error isolation, cloud storage, observability, and repeatable CI/CD-backed releases at scale.
- Reduced dashboard latency from 1.2s to 350ms by optimizing indexed queries and **multi-tenant data models** for an Azure SQL platform supporting 500+ users with row-level isolation controls.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive AI-assisted workflows during concurrent model execution and production traffic reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **maintainable microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments at scale.
- Increased pipeline efficiency by 68% through **AWS Glue and S3 automation**, transforming structured data while reducing storage costs by 15% and supporting repeatable cloud delivery workflows reliably.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **microservice platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with Node.js, WebSockets, and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **real-time data workflows**, integrating automated tests, Docker/Jenkins deployments, Azure Monitor, Sentry, and Kibana across modular production services and release pipelines reliably.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Delivered sub-200ms **Flask APIs** across 50K+ booking records, implementing pricing, coupons, Firebase transactions, and real-time status synchronization for TypeScript-backed mobile and web customer experiences at production scale.
- Reduced manual refund handling by 70% through **reliable webhook automation**, integrating disputes, JWT access, customer requirements, and hourly Databricks/Spark reporting pipelines across secure transactional workflows at scale.

Skills

Languages: Java, Python, C++, SQL, TypeScript, JavaScript, Bash
Backend & Systems: Distributed Systems, Microservices, FastAPI, Flask, Node.js, REST APIs, API Design, Data Modeling, Multithreading
Cloud & Data: AWS, Azure, GCP, Kubernetes, Docker, PostgreSQL, Azure SQL, AWS RDS, AWS S3, MongoDB
Reliability & Security: Fault Isolation, Observability, Performance Tuning, Data Validation, Error Handling, RBAC, JWT, Sentry
Engineering & AI: CI/CD, GitHub Actions, Jenkins, Git, PyTest, Mockito, Unit Testing, Integration Testing, LLM Workflows, AI-Assisted Development

Projects

Secure Media Ingestion Service - Design Concept

- Proposed a **Python cloud ingestion service** validating media and documents, generating checksums, isolating malformed inputs, persisting metadata, enforcing access policies, and supporting idempotent retries with auditable processing states.

AI-Assisted Change Validator - Design Concept

- Proposed an **AI-assisted engineering workflow** reviewing generated code against repository context, running targeted tests and static checks, flagging risky changes, and requiring human approval before merge or deployment.

FinNexus | Live

- Built a **Kubernetes-deployed TypeScript platform** supporting 50K+ users and 1M+ transactions with secure Plaid APIs, MongoDB, WebSockets, token-based controls, real-time synchronization, filtering, and automated delivery workflows reliably.